

51. The average weight of 16 boys in a class is 50.25 kgs and that of the remaining 8 boys is 45.15 kgs. Find the average weight of all the boys in the class.
 (a) 47.55 kgs (b) 48 kgs
 (c) 48.55 kgs (d) 49.25 kgs
52. The mean of 5 observations is 60, the mean of 10 observations is 30 and the mean of 15 observations is 20. The mean of all the 30 observations is (P.C.S., 2009)
 (a) 20 (b) 25
 (c) 30 (d) 40
53. The average expenditure of a man for the first five months of a year is ₹ 5000 and for the next seven months it is ₹ 5400. He saves ₹ 2300 during the year. His average monthly income is (C.P.O., 2007)
 (a) ₹ 5425 (b) ₹ 5446
 (c) ₹ 5500 (d) ₹ 5600
54. A school has 4 sections of Chemistry in Class X having 40, 35, 45 and 42 students. The mean marks obtained in Chemistry test are 50, 60, 55 and 45 respectively for the 4 sections. Determine the overall average of marks per student. (M.A.T., 2006)
 (a) 50.25 (b) 51.25
 (c) 52.25 (d) 53.25
55. If the average of m numbers is n^2 and that of n numbers is m^2 , then the average of $(m + n)$ numbers is (S.S.C., 2008)
 (a) $m - n$ (b) mn
 (c) $m + n$ (d) $\frac{m}{n}$
56. The average of five consecutive numbers is x . If the next two numbers are also included, how shall the average vary? (M.A.T., 2005, C.P.O., 2006, C.P.F. 2008)
 (a) It shall increase by 1
 (b) It shall remain the same
 (c) It shall increase by 1.4
 (d) It shall increase by 2
57. A person purchases 1 kg of tomatoes from each of the 4 places at the rate of 1 kg, 2 kg, 3 kg, 4 kg per rupee respectively. On an average, he has purchased x kg of tomatoes per rupee. Then the value of x is (M.A.T., 2006)
 (a) 1.92 (b) 2
 (c) 2.5 (d) None of these
58. In a certain factory there are five workers A, B, C, D and E. A can complete a work in 4 minutes, B in 5 minutes, C in 6 minutes, D in 10 minutes and E in 12 minutes. The average number of units of work completed per worker per minute will be (P.C.S., 2009)
 (a) 0.16 (b) 0.172
 (c) 0.80 (d) 0.87
59. A car owner buys petrol at ₹ 17, ₹ 19 and ₹ 20 per litre for three consecutive years. Compute the average cost per litre, if he spends ₹ 6460 per year.
 (a) ₹ 18.49 (b) ₹ 18.58
 (c) ₹ 19.2 (d) ₹ 21.66
60. The average of six numbers is x and the average of three of these is y . If the average of the remaining three is z , then
 (a) $x = y + z$ (b) $2x = y + z$
 (c) $x = 2y + 2z$ (d) None of these
61. Out of 9 persons, 8 persons spent ₹ 30 each for their meals. The ninth one spent ₹ 20 more than the average expenditure of all the nine. The total money spent by all of them was
 (a) ₹ 260 (b) ₹ 290
 (c) ₹ 292.50 (d) ₹ 400.50
62. There were 35 students in a hostel. If the number of the students is increased by 7, then the expenses of the mess increase by ₹ 42 per day, while the average expenditure per head diminishes by ₹ 1. The original expenditure of the mess per day was (S.S.C., 2005)
 (a) ₹ 400 (b) ₹ 420
 (c) ₹ 432 (d) ₹ 442
63. The average price of three items of furniture is ₹ 15000. If their prices are in the ratio 3 : 5 : 7, the price of the cheapest item is
 (a) ₹ 9000 (b) ₹ 15000
 (c) ₹ 18000 (d) ₹ 21000
64. Of the four numbers, the first is twice the second, the second is one-third of the third and the third is 5 times the fourth. The average of the numbers is 24.75. The largest of these numbers is
 (a) 9 (b) 25
 (c) 30 (d) None of these
65. Out of three numbers, the first is twice the second and is half of the third. If the average of the three numbers is 56, then difference of first and third numbers is (S.S.C., 2005)
 (a) 12 (b) 20
 (c) 24 (d) 48
66. Of the three numbers, the first is twice the second and the second is twice the third. The average of the reciprocal of the numbers is $\frac{7}{72}$. The numbers are :
 (a) 16, 8, 4 (b) 20, 10, 5
 (c) 24, 12, 6 (d) 36, 18, 9
67. Of the three numbers, the average of the first and the second is greater than the average of the second and the third by 15. What is the difference between the first and the third of the three numbers?
 (a) 15 (b) 45
 (c) 60 (d) Data inadequate
 (e) None of these